

Task 6 : Create a Strong Password and Evaluate Its Strength.

Objective: Understand what makes a password strong and test it against password strength tools.

1. Create multiple passwords.
 - a. password
 - b. sheher234
 - c. iamneo234
 - d. zelenskySays0
 - e. zelenskySays@123#90

The screenshot shows a web browser with the URL passwordmeter.com. The page title is "The Password Meter". Below the title, there is a "Test Your Password" section with a password input field containing "password", a "Hide" checkbox, a "Score" of 8% (indicated by a red bar), and a "Complexity" of "Very Weak". To the right, the "Minimum Requirements" are listed: Minimum 8 characters in length, and Contains 3/4 of the following items: - Uppercase Letters, - Lowercase Letters, - Numbers, - Symbols. Below this, there is a table showing the breakdown of the score.

Test Your Password		Minimum Requirements	
Password:	password	- Minimum 8 characters in length - Contains 3/4 of the following items: - Uppercase Letters - Lowercase Letters - Numbers - Symbols	
Hide:	☐		
Score:	8%		
Complexity:	Very Weak		

Additions		Type	Rate	Count	Bonus
✓	Number of Characters	Flat	$+(n*4)$	8	+ 32
✗	Uppercase Letters	Cond/Incr	$+(len-n)*2$	0	0
✗	Lowercase Letters	Cond/Incr	$+(len-n)*2$	8	0
✗	Numbers	Cond	$+(n*4)$	0	0
✗	Symbols	Flat	$+(n*6)$	0	0
✗	Middle Numbers or Symbols	Flat	$+(n*2)$	0	0
✗	Requirements	Flat	$+(n*2)$	2	0
Deductions					
⚠	Letters Only	Flat	$-n$	8	- 8

Password: password

Score: 8%

Complexity: Very Weak

Feedback: Too short, no symbols, no digits

(110) WhatsApp

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Inbox (3,2)

Password

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The Password Meter

Test Your Password

Password:

Hide: ☐

Score:

39%

Complexity: Weak

Minimum Requirements

- Minimum 8 characters in length
- Contains 3/4 of the following items:
 - Uppercase Letters
 - Lowercase Letters
 - Numbers
 - Symbols

Additions	Type	Rate	Count	Bonus
☒ Number of Characters	Flat	$+(n*4)$	9	+ 36
☒ Uppercase Letters	Cond/Incr	$+\left((len-n)*2\right)$	0	0
☒ Lowercase Letters	Cond/Incr	$+\left((len-n)*2\right)$	6	+ 6
☒ Numbers	Cond	$+(n*4)$	3	+ 12
☒ Symbols	Flat	$+(n*6)$	0	0
☒ Middle Numbers or Symbols	Flat	$+(n*2)$	2	+ 4
☒ Requirements	Flat	$+(n*2)$	3	0

Deductions

☒ Letters Only	Flat	$-n$	0	0
☒ Numbers Only	Flat	$-n$	0	0

Password: sheher234

Score: 39%

Complexity: Weak

Feedback: Add special characters

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Inbox (3,2)

Password

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The Password Meter

Test Your Password

Password:

Hide: ☐

Score:

41%

Complexity: Good

Minimum Requirements

- Minimum 8 characters in length
- Contains 3/4 of the following items:
 - Uppercase Letters
 - Lowercase Letters
 - Numbers
 - Symbols

Additions	Type	Rate	Count	Bonus
☒ Number of Characters	Flat	$+(n*4)$	9	+ 36
☒ Uppercase Letters	Cond/Incr	$+\left((len-n)*2\right)$	0	0
☒ Lowercase Letters	Cond/Incr	$+\left((len-n)*2\right)$	6	+ 6
☒ Numbers	Cond	$+(n*4)$	3	+ 12
☒ Symbols	Flat	$+(n*6)$	0	0
☒ Middle Numbers or Symbols	Flat	$+(n*2)$	2	+ 4
☒ Requirements	Flat	$+(n*2)$	3	0

Deductions

☒ Letters Only	Flat	$-n$	0	0
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Password: iamneo234

Score: 41%

Complexity: Good

Feedback: add Special characters

(110) What's up

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Inbox (3,2)

Password

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The Password Meter

Test Your Password

Password:

zelenskySays0

Hide:

☐

Score:

73%

Complexity:

Strong

Minimum Requirements

- Minimum 8 characters in length
- Contains 3/4 of the following items:
 - Uppercase Letters
 - Lowercase Letters
 - Numbers
 - Symbols

Additions	Type	Rate	Count	Bonus
Number of Characters	Flat	$+(n*4)$	13	+ 52
Uppercase Letters	Cond/Incr	$+\left((len-n)*2\right)$	1	+ 24
Lowercase Letters	Cond/Incr	$+\left((len-n)*2\right)$	11	+ 4
Numbers	Cond	$+(n*4)$	1	+ 4
Symbols	Flat	$+(n*6)$	0	0
Middle Numbers or Symbols	Flat	$+(n*2)$	0	0
Requirements	Flat	$+(n*2)$	4	+ 8

Deductions

Letters Only	Flat	$-n$	0	0
Numbers Only	Flat	$-n$	0	0

Password: zelenskySays0

Score: 73%

Complexity: Strong

Feedback: Good length

(110) What's up

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Inbox (3,2)

Password

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The Password Meter

Test Your Password

Password:

zelenskySays@123#90

Hide:

☐

Score:

100%

Complexity:

Very Strong

Minimum Requirements

- Minimum 8 characters in length
- Contains 3/4 of the following items:
 - Uppercase Letters
 - Lowercase Letters
 - Numbers
 - Symbols

Additions	Type	Rate	Count	Bonus
Number of Characters	Flat	$+(n*4)$	19	+ 76
Uppercase Letters	Cond/Incr	$+\left((len-n)*2\right)$	1	+ 36
Lowercase Letters	Cond/Incr	$+\left((len-n)*2\right)$	11	+ 16
Numbers	Cond	$+(n*4)$	5	+ 20
Symbols	Flat	$+(n*6)$	2	+ 12
Middle Numbers or Symbols	Flat	$+(n*2)$	6	+ 12
Requirements	Flat	$+(n*2)$	5	+ 10

Deductions

Letters Only	Flat	$-n$	0	0
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Password: zelenskySays@123#90

Score: 100%

Complexity: Very Strong.

Feedback: Good length and enough complexity.

Tips for Creating Strong Passwords

1. **Use at least 12–16 characters** for better security.
2. **Include a mix of uppercase, lowercase, numbers, and symbols** to increase complexity.
3. **Avoid common words, names, or predictable patterns** like Password123 or qwerty.
4. **Do not reuse passwords** across multiple accounts.
5. **Randomize characters** rather than using easily guessable substitutions (e.g., H3llo@123 is still predictable).
6. **Avoid personal information** (name, birthdate, phone number).
7. **Use a passphrase with random words** (e.g., Horse!Cloud7River@) for better memorability.
8. **Enable Two-Factor Authentication (2FA)** wherever possible for added security.
9. **Consider a password manager** for generating and storing strong, unique passwords.
10. **Change passwords periodically**, especially after a suspected breach.

Common Password Attacks

1. **Brute Force Attack**
 - Tries every possible combination of characters until the correct password is found.
 - Stronger passwords with length and complexity take exponentially longer to crack.
2. **Dictionary Attack**
 - Uses a list of common words or leaked passwords to guess the password.
 - Passwords like password123 or hello123 are easily cracked this way.
3. **Credential Stuffing**
 - Attackers use leaked username-password combinations from one site on other sites, exploiting reused passwords.
4. **Phishing Attack**
 - Tricks users into revealing passwords via fake login pages, emails, or messages.
5. **Keylogging**
 - Malware records keystrokes to steal passwords as you type them.
6. **Rainbow Table Attack**
 - Uses precomputed hash values to crack passwords quickly if they are not salted.
7. **Social Engineering**
 - Manipulates or tricks users into sharing passwords (e.g., pretending to be IT support).

How Password Complexity Affects Security

1. Increases Brute Force Time

- A short password (e.g., abc123) can be cracked in **seconds**.
- A complex 12+ character password with mixed symbols can take **years or centuries** to crack using brute force.

2. Prevents Dictionary Attacks

- Simple words or patterns are easy to guess from common wordlists.
- Adding random symbols, numbers, and case sensitivity makes dictionary attacks ineffective.

3. Reduces Predictability

- Attackers often try patterns like Name@123 or Password!.
- Complexity and randomness eliminate predictable sequences.

4. Strengthens Against Rainbow Tables

- Complex passwords combined with salting make precomputed hash tables useless.

5. Mitigates Credential Stuffing

- Unique, complex passwords for every account ensure that a breach on one site doesn't compromise others.